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# An Introduction to the Bootstrap

Bradley Efron
Robert J. Tibshirani

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# An Introduction to the Bootstrap

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*AND TO THE MEMORY OF*

*RUPERT G. MILLER, JR.*



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# Contents


**Preface** **xiv**

**1 Introduction** **1**
1.1 An overview of this book 6
1.2 Information for instructors 8
1.3 Some of the notation used in the book 9

**2 The accuracy of a sample mean** **10**
2.1 Problems 15

**3 Random samples and probabilities** **17**
3.1 Introduction 17
3.2 Random samples 17
3.3 Probability theory 20
3.4 Problems 28

**4 The empirical distribution function and the plug-in principle** **31**
4.1 Introduction 31
4.2 The empirical distribution function 31
4.3 The plug-in principle 35
4.4 Problems 37

**5 Standard errors and estimated standard errors** **39**
5.1 Introduction 39
5.2 The standard error of a mean 39
5.3 Estimating the standard error of the mean 42
5.4 Problems 43

**6 The bootstrap estimate of standard error** **45**
6.1 Introduction 45
6.2 The bootstrap estimate of standard error 45
6.3 Example: the correlation coefficient 49
6.4 The number of bootstrap replications $B$ 50
6.5 The parametric bootstrap 53
6.6 Bibliographic notes 56
6.7 Problems 57

**7 Bootstrap standard errors: some examples** **60**
7.1 Introduction 60
7.2 Example 1: test score data 61
7.3 Example 2: curve fitting 70
7.4 An example of bootstrap failure 81
7.5 Bibliographic notes 81
7.6 Problems 82

**8 More complicated data structures** **86**
8.1 Introduction 86
8.2 One-sample problems 86
8.3 The two-sample problem 88
8.4 More general data structures 90
8.5 Example: lutenizing hormone 92
8.6 The moving blocks bootstrap 99
8.7 Bibliographic notes 102
8.8 Problems 103

**9 Regression models** **105**
9.1 Introduction 105
9.2 The linear regression model 105
9.3 Example: the hormone data 107
9.4 Application of the bootstrap 111
9.5 Bootstrapping pairs vs bootstrapping residuals 113
9.6 Example: the cell survival data 115
9.7 Least median of squares 117
9.8 Bibliographic notes 121
9.9 Problems 121

**10 Estimates of bias** **124**
10.1 Introduction 124

10.2 The bootstrap estimate of bias 124
10.3 Example: the patch data 126
10.4 An improved estimate of bias 130
10.5 The jackknife estimate of bias 133
10.6 Bias correction 138
10.7 Bibliographic notes 139
10.8 Problems 139

**11 The jackknife 141**
11.1 Introduction 141
11.2 Definition of the jackknife 141
11.3 Example: test score data 143
11.4 Pseudo-values 145
11.5 Relationship between the jackknife and bootstrap 145
11.6 Failure of the jackknife 148
11.7 The delete-$d$ jackknife 149
11.8 Bibliographic notes 149
11.9 Problems 150

**12 Confidence intervals based on bootstrap "tables" 153**
12.1 Introduction 153
12.2 Some background on confidence intervals 155
12.3 Relation between confidence intervals and hypothesis tests 156
12.4 Student's $t$ interval 158
12.5 The bootstrap-$t$ interval 160
12.6 Transformations and the bootstrap-$t$ 162
12.7 Bibliographic notes 166
12.8 Problems 166

**13 Confidence intervals based on bootstrap percentiles 168**
13.1 Introduction 168
13.2 Standard normal intervals 168
13.3 The percentile interval 170
13.4 Is the percentile interval backwards? 174
13.5 Coverage performance 174
13.6 The transformation-respecting property 175
13.7 The range-preserving property 176
13.8 Discussion 176

13.9 Bibliographic notes 176
13.10 Problems 177

**14 Better bootstrap confidence intervals 178**
14.1 Introduction 178
14.2 Example: the spatial test data 179
14.3 The $BC_a$ method 184
14.4 The ABC method 188
14.5 Example: the tooth data 190
14.6 Bibliographic notes 199
14.7 Problems 199

**15 Permutation tests 202**
15.1 Introduction 202
15.2 The two-sample problem 202
15.3 Other test statistics 210
15.4 Relationship of hypothesis tests to confidence intervals and the bootstrap 214
15.5 Bibliographic notes 218
15.6 Problems 218

**16 Hypothesis testing with the bootstrap 220**
16.1 Introduction 220
16.2 The two-sample problem 220
16.3 Relationship between the permutation test and the bootstrap 223
16.4 The one-sample problem 224
16.5 Testing multimodality of a population 227
16.6 Discussion 232
16.7 Bibliographic notes 233
16.8 Problems 234

**17 Cross-validation and other estimates of prediction error 237**
17.1 Introduction 237
17.2 Example: hormone data 238
17.3 Cross-validation 239
17.4 $C_p$ and other estimates of prediction error 242
17.5 Example: classification trees 243
17.6 Bootstrap estimates of prediction error 247

17.6.1 Overview 247
17.6.2 Some details 249
17.7 The .632 bootstrap estimator 252
17.8 Discussion 254
17.9 Bibliographic notes 255
17.10 Problems 255

**18 Adaptive estimation and calibration 258**
18.1 Introduction 258
18.2 Example: smoothing parameter selection for curve fitting 258
18.3 Example: calibration of a confidence point 263
18.4 Some general considerations 266
18.5 Bibliographic notes 268
18.6 Problems 269

**19 Assessing the error in bootstrap estimates 271**
19.1 Introduction 271
19.2 Standard error estimation 272
19.3 Percentile estimation 273
19.4 The jackknife-after-bootstrap 275
19.5 Derivations 280
19.6 Bibliographic notes 281
19.7 Problems 281

**20 A geometrical representation for the bootstrap and jackknife 283**
20.1 Introduction 283
20.2 Bootstrap sampling 285
20.3 The jackknife as an approximation to the bootstrap 287
20.4 Other jackknife approximations 289
20.5 Estimates of bias 290
20.6 An example 293
20.7 Bibliographic notes 295
20.8 Problems 295

**21 An overview of nonparametric and parametric inference 296**
21.1 Introduction 296
21.2 Distributions, densities and likelihood functions 296

21.3 Functional statistics and influence functions 298
21.4 Parametric maximum likelihood inference 302
21.5 The parametric bootstrap 306
21.6 Relation of parametric maximum likelihood, bootstrap and jackknife approaches 307
21.6.1 Example: influence components for the mean 309
21.7 The empirical cdf as a maximum likelihood estimate 310
21.8 The sandwich estimator 310
21.8.1 Example: Mouse data 311
21.9 The delta method 313
21.9.1 Example: delta method for the mean 315
21.9.2 Example: delta method for the correlation coefficient 315
21.10 Relationship between the delta method and infinitesimal jackknife 315
21.11 Exponential families 316
21.12 Bibliographic notes 319
21.13 Problems 320

**22 Further topics in bootstrap confidence intervals 321**
22.1 Introduction 321
22.2 Correctness and accuracy 321
22.3 Confidence points based on approximate pivots 322
22.4 The $BC_a$ interval 325
22.5 The underlying basis for the $BC_a$ interval 326
22.6 The ABC approximation 328
22.7 Least favorable families 331
22.8 The $ABC_q$ method and transformations 333
22.9 Discussion 334
22.10 Bibliographic notes 335
22.11 Problems 335

**23 Efficient bootstrap computations 338**
23.1 Introduction 338
23.2 Post-sampling adjustments 340
23.3 Application to bootstrap bias estimation 342
23.4 Application to bootstrap variance estimation 346
23.5 Pre- and post-sampling adjustments 348
23.6 Importance sampling for tail probabilities 349
23.7 Application to bootstrap tail probabilities 352

23.8 Bibliographic notes 356
23.9 Problems 357

**24 Approximate likelihoods 358**
24.1 Introduction 358
24.2 Empirical likelihood 360
24.3 Approximate pivot methods 362
24.4 Bootstrap partial likelihood 364
24.5 Implied likelihood 367
24.6 Discussion 370
24.7 Bibliographic notes 371
24.8 Problems 371

**25 Bootstrap bioequivalence 372**
25.1 Introduction 372
25.2 A bioequivalence problem 372
25.3 Bootstrap confidence intervals 374
25.4 Bootstrap power calculations 379
25.5 A more careful power calculation 381
25.6 Fieller's intervals 384
25.7 Bibliographic notes 389
25.8 Problems 389

**26 Discussion and further topics 392**
26.1 Discussion 392
26.2 Some questions about the bootstrap 394
26.3 References on further topics 396

**Appendix: software for bootstrap computations 398**
Introduction 398
Some available software 399
S language functions 399

**References 413**

**Author index 426**

**Subject index 430**

# Preface

*Dear friend, theory is all gray,*
*and the golden tree of life is green.*
Goethe, from "Faust"

*The ability to simplify means to eliminate the unnecessary so that the necessary may speak.*
Hans Hoffmann

Statistics is a subject of amazingly many uses and surprisingly few effective practitioners. The traditional road to statistical knowledge is blocked, for most, by a formidable wall of mathematics. Our approach here avoids that wall. The bootstrap is a computer-based method of statistical inference that can answer many real statistical questions without formulas. Our goal in this book is to arm scientists and engineers, as well as statisticians, with computational techniques that they can use to analyze and understand complicated data sets.

The word "understand" is an important one in the previous sentence. This is not a statistical cookbook. We aim to give the reader a good intuitive understanding of statistical inference.

One of the charms of the bootstrap is the direct appreciation it gives of variance, bias, coverage, and other probabilistic phenomena. What does it mean that a confidence interval contains the true value with probability .90? The usual textbook answer appears formidably abstract to most beginning students. Bootstrap confidence intervals are directly constructed from real data sets, using a simple computer algorithm. This doesn't necessarily make it easy to understand confidence intervals, but at least the difficulties are the appropriate conceptual ones, and not mathematical muddles.

Much of the exposition in our book is based on the analysis of real data sets. The mouse data, the stamp data, the tooth data, the hormone data, and other small but genuine examples, are an important part of the presentation. These are especially valuable if the reader can try his own computations on them. Personal computers are sufficient to handle most bootstrap computations for these small data sets.

This book does not give a rigorous technical treatment of the bootstrap, and we concentrate on the ideas rather than their mathematical justification. Many of these ideas are quite sophisticated, however, and this book is not just for beginners. The presentation starts off slowly but builds in both its scope and depth. More mathematically advanced accounts of the bootstrap may be found in papers and books by many researchers that are listed in the Bibliographic notes at the end of the chapters.

We would like to thank Andreas Buja, Anthony Davison, Peter Hall, Trevor Hastie, John Rice, Bernard Silverman, James Stafford and Sami Tibshirani for making very helpful comments and suggestions on the manuscript. We especially thank Timothy Hesterberg and Cliff Lunneborg for the great deal of time and effort that they spent on reading and preparing comments. Thanks to Maria-Luisa Gardner for providing expert advice on the "rules of punctuation." We would also like to thank numerous students at both Stanford University and the University of Toronto for pointing out errors in earlier drafts, and colleagues and staff at our universities for their support. Thanks to Tom Glinos of the University of Toronto for maintaining a healthy computing environment. Karola DeCleve typed much of the first draft of this book, and maintained vigilance against errors during its entire history. All of this was done cheerfully and in a most helpful manner, for which we are truly grateful. Trevor Hastie provided expert "S" and TEX advice, at crucial stages in the project.

We were lucky to have not one but two superb editors working on this project. Bea Schube got us going, before starting her retirement; Bea has done a great deal for the statistics profession and we wish her all the best. John Kimmel carried the ball after Bea left, and did an excellent job. We thank our copy-editor Jim Geronimo for his thorough correction of the manuscript, and take responsibility for any errors that remain.

The first author was supported by the National Institutes of Health and the National Science Foundation. Both groups have

supported the development of statistical theory at Stanford, including much of the theory behind this book. The second author would like to thank his wife Cheryl for her understanding and support during this entire project, and his parents for a lifetime of encouragement. He gratefully acknowledges the support of the Natural Sciences and Engineering Research Council of Canada.

*Palo Alto and Toronto* Bradley Efron
*June 1993* Robert Tibshirani

CHAPTER 1

# Introduction

Statistics is the science of learning from experience, especially experience that arrives a little bit at a time. The earliest information science was statistics, originating in about 1650. This century has seen statistical techniques become the analytic methods of choice in biomedical science, psychology, education, economics, communications theory, sociology, genetic studies, epidemiology, and other areas. Recently, traditional sciences like geology, physics, and astronomy have begun to make increasing use of statistical methods as they focus on areas that demand informational efficiency, such as the study of rare and exotic particles or extremely distant galaxies.

Most people are not natural-born statisticians. Left to our own devices we are not very good at picking out patterns from a sea of noisy data. To put it another way, we are all too good at picking out non-existent patterns that happen to suit our purposes. Statistical theory attacks the problem from both ends. It provides optimal methods for finding a real signal in a noisy background, and also provides strict checks against the overinterpretation of random patterns.

Statistical theory attempts to answer three basic questions:

(1) How should I collect my data?

(2) How should I analyze and summarize the data that I've collected?

(3) How accurate are my data summaries?

Question 3 constitutes part of the process known as statistical inference. The bootstrap is a recently developed technique for making certain kinds of statistical inferences. It is only recently developed because it requires modern computer power to simplify the often intricate calculations of traditional statistical theory.

The explanations that we will give for the bootstrap, and other

computer-based methods, involve explanations of traditional ideas in statistical inference. The basic ideas of statistics haven't changed, but their implementation has. The modern computer lets us apply these ideas flexibly, quickly, easily, and with a minimum of mathematical assumptions. Our primary purpose in the book is to explain when and why bootstrap methods work, and how they can be applied in a wide variety of real data-analytic situations.

All three basic statistical concepts, data collection, summary and inference, are illustrated in the New York Times excerpt of Figure 1.1. A study was done to see if small aspirin doses would prevent heart attacks in healthy middle-aged men. The data for the aspirin study were collected in a particularly efficient way: by a controlled, randomized, double-blind study. One half of the subjects received aspirin and the other half received a control substance, or placebo, with no active ingredients. The subjects were randomly assigned to the aspirin or placebo groups. Both the subjects and the supervising physicians were blinded to the assignments, with the statisticians keeping a secret code of who received which substance. Scientists, like everyone else, want the project they are working on to succeed. The elaborate precautions of a controlled, randomized, blinded experiment guard against seeing benefits that don't exist, while maximizing the chance of detecting a genuine positive effect.

The summary statistics in the newspaper article are very simple:

| | heart attacks (fatal plus non-fatal) | subjects |
|---|---|---|
| aspirin group: | 104 | 11037 |
| placebo group: | 189 | 11034 |

We will see examples of much more complicated summaries in later chapters. One advantage of using a good experimental design is a simplification of its results. What strikes the eye here is the lower rate of heart attacks in the aspirin group. The ratio of the two rates is

$$\widehat{\theta} = \frac{104/11037}{189/11034} = .55. \tag{1.1}$$

If this study can be believed, and its solid design makes it very believable, the aspirin-takers only have 55% as many heart attacks as placebo-takers.

Of course we are not really interested in $\widehat{\theta}$, the estimated ratio. What we would like to know is $\theta$, the true ratio, that is the ratio

# HEART ATTACK RISK FOUND TO BE CUT BY TAKING ASPIRIN

## LIFESAVING EFFECTS SEEN

## Study Finds Benefit of Tablet Every Other Day Is Much Greater Than Expected

By HAROLD M. SCHMECK Jr.

A major nationwide study shows that a single aspirin tablet every other day can sharply reduce a man's risk of heart attack and death from heart attack.

The lifesaving effects were so dramatic that the study was halted in mid-December so that the results could be reported as soon as possible to the participants and to the medical profession in general.

The magnitude of the beneficial effect was far greater than expected, Dr. Charles H. Hennekens of Harvard, principal investigator in the research, said in a telephone interview. The risk of myocardial infarction, the technical name for heart attack, was cut almost in half.

**'Extreme Beneficial Effect'**

A special report said the results showed "a statistically extreme beneficial effect" from the use of aspirin. The report is to be published Thursday in The New England Journal of Medicine.

In recent years smaller studies have demonstrated that a person who has had one heart attack can reduce the risk of a second by taking aspirin, but there had been no proof that the beneficial effect would extend to the general male population.

Dr. Claude Lenfant, the director of the National Heart Lung and Blood Institute, said the findings were "extremely important," but he said the general public should not take the report as an indication that everyone should start taking aspirin.

Figure 1.1. *Front-page news from the New York Times of January 27, 1987. Reproduced by permission of the New York Times.*

we would see if we could treat all subjects, and not just a sample of them. The value $\widehat{\theta} = .55$ is only an estimate of $\theta$. The sample seems large here, 22071 subjects in all, but the conclusion that aspirin works is really based on a smaller number, the 293 observed heart attacks. How do we know that $\widehat{\theta}$ might not come out much less favorably if the experiment were run again?

This is where statistical inference comes in. Statistical theory allows us to make the following inference: the true value of $\theta$ lies in the interval

$$.43 < \theta < .70 \tag{1.2}$$

with 95% confidence. Statement (1.2) is a classical confidence interval, of the type discussed in Chapters 12–14, and 22. It says that if we ran a much bigger experiment, with millions of subjects, the ratio of rates probably wouldn't be too much different than (1.1). We almost certainly wouldn't decide that $\theta$ exceeded 1, that is that aspirin was actually harmful. It is really rather amazing that the same data that give us an estimated value, $\widehat{\theta} = .55$ in this case, also can give us a good idea of the estimate's accuracy.

Statistical inference is serious business. A lot can ride on the decision of whether or not an observed effect is real. The aspirin study tracked strokes as well as heart attacks, with the following results:

| | strokes | subjects |
|---|---|---|
| aspirin group: | 119 | 11037 |
| placebo group: | 98 | 11034 |

(1.3)

For strokes, the ratio of rates is

$$\widehat{\theta} = \frac{119/11037}{98/11034} = 1.21. \tag{1.4}$$

It now looks like taking aspirin is actually harmful. However the interval for the true stroke ratio $\theta$ turns out to be

$$.93 < \theta < 1.59 \tag{1.5}$$

with 95% confidence. This includes the neutral value $\theta = 1$, at which aspirin would be no better or worse than placebo vis-à-vis strokes. In the language of statistical hypothesis testing, aspirin was found to be significantly beneficial for preventing heart attacks, but not significantly harmful for causing strokes. The opposite conclusion had been reached in an older, smaller study concerning men

who had experienced previous heart attacks. The aspirin treatment remains mildly controversial for such patients.

The bootstrap is a data-based simulation method for statistical inference, which can be used to produce inferences like (1.2) and (1.5). The use of the term bootstrap derives from the phrase *to pull oneself up by one's bootstrap*, widely thought to be based on one of the eighteenth century Adventures of Baron Munchausen, by Rudolph Erich Raspe. (The Baron had fallen to the bottom of a deep lake. Just when it looked like all was lost, he thought to pick himself up by his own bootstraps.) It is not the same as the term "bootstrap" used in computer science meaning to "boot" a computer from a set of core instructions, though the derivation is similar.

Here is how the bootstrap works in the stroke example. We create two populations: the first consisting of 119 ones and 11037-119=10918 zeroes, and the second consisting of 98 ones and 11034-98=10936 zeroes. We draw with replacement a sample of 11037 items from the first population, and a sample of 11034 items from the second population. Each of these is called a *bootstrap sample.* From these we derive the bootstrap replicate of $\hat{\theta}$:

$$\hat{\theta}^* = \frac{\text{Proportion of ones in bootstrap sample \#1}}{\text{Proportion of ones in bootstrap sample \#2}}. \tag{1.6}$$

We repeat this process a large number of times, say 1000 times, and obtain 1000 *bootstrap replicates* $\hat{\theta}^*$. This process is easy to implement on a computer, as we will see later. These 1000 replicates contain information that can be used to make inferences from our data. For example, the standard deviation turned out to be 0.17 in a batch of 1000 replicates that we generated. The value 0.17 is an estimate of the standard error of the ratio of rates $\hat{\theta}$. This indicates that the observed ratio $\hat{\theta} = 1.21$ is only a little more than one standard error larger than 1, and so the neutral value $\theta = 1$ cannot be ruled out. A rough 95% confidence interval like (1.5) can be derived by taking the 25th and 975th largest of the 1000 replicates, which in this case turned out to be (.93, 1.60).

In this simple example, the confidence interval derived from the bootstrap agrees very closely with the one derived from statistical theory. Bootstrap methods are intended to simplify the calculation of inferences like (1.2) and (1.5), producing them in an automatic way even in situations much more complicated than the aspirin study.

The terminology of statistical summaries and inferences, like regression, correlation, analysis of variance, discriminant analysis, standard error, significance level and confidence interval, has become the lingua franca of all disciplines that deal with noisy data. We will be examining what this language means and how it works in practice. The particular goal of bootstrap theory is a computer-based implementation of basic statistical concepts. In some ways it is easier to understand these concepts in computer-based contexts than through traditional mathematical exposition.

## 1.1 An overview of this book

This book describes the bootstrap and other methods for assessing statistical accuracy. The bootstrap does not work in isolation but rather is applied to a wide variety of statistical procedures. Part of the objective of this book is expose the reader to many exciting and useful statistical techniques through real-data examples. Some of the techniques described include nonparametric regression, density estimation, classification trees, and least median of squares regression.

Here is a chapter-by-chapter synopsis of the book. **Chapter 2** introduces the bootstrap estimate of standard error for a simple mean. **Chapters 3–5** contain some basic background material, and may be skimmed by readers eager to get to the details of the bootstrap in **Chapter 6**. Random samples, populations, and basic probability theory are reviewed in **Chapter 3**. **Chapter 4** defines the empirical distribution function estimate of the population, which simply estimates the probability of each of $n$ data items to be $1/n$. **Chapter 4** also shows that many familiar statistics can be viewed as "plug-in" estimates, that is, estimates obtained by plugging in the empirical distribution function for the unknown distribution of the population. **Chapter 5** reviews standard error estimation for a mean, and shows how the usual textbook formula can be derived as a simple plug-in estimate.

The bootstrap is defined in **Chapter 6**, for estimating the standard error of a statistic from a single sample. The bootstrap standard error estimate is a plug-in estimate that rarely can be computed exactly; instead a simulation ("resampling") method is used for approximating it.

**Chapter 7** describes the application of bootstrap standard errors in two complicated examples: a principal components analysis

and a curve fitting problem.

Up to this point, only one-sample data problems have been discussed. The application of the bootstrap to more complicated data structures is discussed in **Chapter 8**. A two-sample problem and a time-series analysis are described.

Regression analysis and the bootstrap are discussed and illustrated in **Chapter 9**. The bootstrap estimate of standard error is applied in a number of different ways and the results are discussed in two examples.

The use of the bootstrap for estimation of bias is the topic of **Chapter 10**, and the pros and cons of bias correction are discussed. **Chapter 11** describes the jackknife method in some detail. We see that the jackknife is a simple closed-form approximation to the bootstrap, in the context of standard error and bias estimation.

The use of the bootstrap for construction of confidence intervals is described in **Chapters 12, 13** and **14**. There are a number of different approaches to this important topic and we devote quite a bit of space to them. In **Chapter 12** we discuss the bootstrap-$t$ approach, which generalizes the usual Student's $t$ method for constructing confidence intervals. The percentile method (**Chapter 13**) uses instead the percentiles of the bootstrap distribution to define confidence limits. The $BC_a$ (bias-corrected accelerated interval) makes important corrections to the percentile interval and is described in **Chapter 14**.

**Chapter 15** covers permutation tests, a time-honored and useful set of tools for hypothesis testing. Their close relationship with the bootstrap is discussed; **Chapter 16** shows how the bootstrap can be used in more general hypothesis testing problems.

Prediction error estimation arises in regression and classification problems, and we describe some approaches for it in **Chapter 17**. Cross-validation and bootstrap methods are described and illustrated. Extending this idea, **Chapter 18** shows how the bootstrap and cross-validation can be used to adapt estimators to a set of data.

Like any statistic, bootstrap estimates are random variables and so have inherent error associated with them. When using the bootstrap for making inferences, it is important to get an idea of the magnitude of this error. In **Chapter 19** we discuss the jackknife-after-bootstrap method for estimating the standard error of a bootstrap quantity.

**Chapters 20–25** contain more advanced material on selected

topics, and delve more deeply into some of the material introduced in the previous chapters. The relationship between the bootstrap and jackknife is studied via the "resampling picture" in **Chapter 20. Chapter 21** gives an overview of non-parametric and parametric inference, and relates the bootstrap to a number of other techniques for estimating standard errors. These include the delta method, Fisher information, infinitesimal jackknife, and the sandwich estimator.

Some advanced topics in bootstrap confidence intervals are discussed in **Chapter 22**, providing some of the underlying basis for the techniques introduced in Chapters 12–14. **Chapter 23** describes methods for efficient computation of bootstrap estimates including control variates and importance sampling. In **Chapter 24** the construction of approximate likelihoods is discussed. The bootstrap and other related methods are used to construct a "non-parametric" likelihood in situations where a parametric model is not specified.

**Chapter 25** describes in detail a bioequivalence study in which the bootstrap is used to estimate power and sample size. In **Chapter 26** we discuss some general issues concerning the bootstrap and its role in statistical inference.

Finally, the **Appendix** contains a description of a number of different computer programs for the methods discussed in this book.

## 1.2 Information for instructors

We envision that this book can provide the basis for (at least) two different one semester courses. An upper-year undergraduate or first-year graduate course could be taught from some or all of the first 19 chapters, possibly covering Chapter 25 as well (both authors have done this). In addition, a more advanced graduate course could be taught from a selection of Chapters 6–19, and a selection of Chapters 20–26. For an advanced course, supplementary material might be used, such as Peter Hall's book *The Bootstrap and Edgeworth Expansion* or journal papers on selected technical topics. The Bibliographic notes in the book contain many suggestions for background reading.

We have provided numerous exercises at the end of each chapter. Some of these involve computing, since it is important for the student to get hands-on experience for learning the material. The bootstrap is most effectively used in a high-level language for data

analysis and graphics. Our language of choice (at present) is "S" (or "S-PLUS"), and a number of S programs appear in the Appendix. Most of these programs could be easily translated into other languages such as Gauss, Lisp-Stat, or Matlab. Details on the availability of S and S-PLUS are given in the Appendix.

## 1.3 Some of the notation used in the book

Lower case bold letters such as $\mathbf{x}$ refer to vectors, that is, $\mathbf{x} = (x_1, x_2, \ldots x_n)$. Matrices are denoted by upper case bold letters such as $\mathbf{X}$, while a plain uppercase letter like $X$ refers to a random variable. The transpose of a vector is written as $\mathbf{x}^T$. A superscript "*" indicates a bootstrap random variable: for example, $\mathbf{x}^*$ indicates a bootstrap data set generated from a data set $\mathbf{x}$. Parameters are denoted by Greek letters such as $\theta$. A hat on a letter indicates an estimate, such as $\hat{\theta}$. The letters $F$ and $G$ refer to populations. In Chapter 21 the same symbols are used for the cumulative distribution function of a population. $I_C$ is the indicator function equal to 1 if condition $C$ is true and 0 otherwise. For example, $I_{\{x<2\}} = 1$ if $x < 2$ and 0 otherwise. The notation $\mathrm{tr}(A)$ refers to the trace of the matrix $A$, that is, the sum of the diagonal elements. The derivatives of a function $g(x)$ are denoted by $g'(x), g''(x)$ and so on.

The notation

$$F \rightarrow (x_1, x_2, \ldots x_n)$$

indicates an independent and identically distributed sample drawn from $F$. Equivalently, we also write $x_i \overset{\text{i.i.d.}}{\sim} F$ for $i = 1, 2, \ldots n$.

Notation such as $\#\{x_i > 3\}$ means the number of $x_i$s greater than 3. $\log x$ refers to the natural logarithm of $x$.

CHAPTER 2

# The accuracy of a sample mean

The bootstrap is a computer-based method for assigning measures of accuracy to statistical estimates. The basic idea behind the bootstrap is very simple, and goes back at least two centuries. After reviewing some background material, this book describes the bootstrap method, its implementation on the computer, and its application to some real data analysis problems. First though, this chapter focuses on the one example of a statistical estimator where we really don't need a computer to assess accuracy: the sample mean. In addition to previewing the bootstrap, this gives us a chance to review some fundamental ideas from elementary statistics. We begin with a simple example concerning means and their estimated accuracies.

Table 2.1 shows the results of a small experiment, in which 7 out of 16 mice were randomly selected to receive a new medical treatment, while the remaining 9 were assigned to the non-treatment (control) group. The treatment was intended to prolong survival after a test surgery. The table shows the survival time following surgery, in days, for all 16 mice.

Did the treatment prolong survival? A comparison of the means for the two groups offers preliminary grounds for optimism. Let $x_1, x_2, \cdots, x_7$ indicate the lifetimes in the treatment group, so $x_1 = 94, x_2 = 197, \cdots, x_7 = 23$, and likewise let $y_1, y_2, \cdots, y_9$ indicate the control group lifetimes. The group means are

$$\bar{x} = \sum_{i=1}^{7} x_i/7 = 86.86 \quad \text{and} \quad \bar{y} = \sum_{i=1}^{9} y_i/9 = 56.22, \tag{2.1}$$

so the difference $\bar{x} - \bar{y}$ equals 30.63, suggesting a considerable life-prolonging effect for the treatment.

But how accurate are these estimates? After all, the means (2.1) are based on small samples, only 7 and 9 mice, respectively. In

Table 2.1. *The mouse data. Sixteen mice were randomly assigned to a treatment group or a control group. Shown are their survival times, in days, following a test surgery. Did the treatment prolong survival?*

| Group | Data | | | (Sample Size) | Mean | Estimated Standard Error |
|---|---|---|---|---|---|---|
| Treatment: | 94 | 197 | 16 | | | |
| | 38 | 99 | 141 | | | |
| | 23 | | | (7) | 86.86 | 25.24 |
| Control: | 52 | 104 | 146 | | | |
| | 10 | 51 | 30 | | | |
| | 40 | 27 | 46 | (9) | 56.22 | 14.14 |
| | | | | Difference: | 30.63 | 28.93 |

order to answer this question, we need an estimate of the accuracy of the sample means $\bar{x}$ and $\bar{y}$. For sample means, and essentially *only* for sample means, an accuracy formula is easy to obtain.

The *estimated standard error* of a mean $\bar{x}$ based on $n$ independent data points $x_1, x_2, \cdots, x_n$, $\bar{x} = \sum_{i=1}^{n} x_i/n$, is given by the formula

$$\sqrt{\frac{s^2}{n}} \tag{2.2}$$

where $s^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2/(n-1)$. (This formula, and standard errors in general, are discussed more carefully in Chapter 5.) The standard error of any estimator is defined to be the square root of its variance, that is, the estimator's root mean square variability around its expectation. This is the most common measure of an estimator's accuracy. Roughly speaking, an estimator will be less than one standard error away from its expectation about 68% of the time, and less than two standard errors away about 95% of the time.

If the estimated standard errors in the mouse experiment were very small, say less than 1, then we would know that $\bar{x}$ and $\bar{y}$ were close to their expected values, and that the observed difference of 30.63 was probably a good estimate of the true survival-prolonging

capability of the treatment. On the other hand, if formula (2.2) gave big estimated standard errors, say 50, then the difference estimate would be too inaccurate to depend on.

The actual situation is shown at the right of Table 2.1. The estimated standard errors, calculated from (2.2), are 25.24 for $\bar{x}$ and 14.14 for $\bar{y}$. The standard error for the difference $\bar{x} - \bar{y}$ equals $28.93 = \sqrt{25.24^2 + 14.14^2}$ (since the variance of the difference of two independent quantities is the sum of their variances). We see that the observed difference 30.63 is only $30.63/28.93 = 1.05$ estimated standard errors greater than zero. Readers familiar with hypothesis testing theory will recognize this as an *insignificant* result, one that could easily arise by chance even if the treatment really had no effect at all.

There are more precise ways to verify this disappointing result, (e.g. the permutation test of Chapter 15), but usually, as in this case, estimated standard errors are an excellent first step toward thinking critically about statistical estimates. Unfortunately standard errors have a major disadvantage: for most statistical estimators other than the mean there is no formula like (2.2) to provide estimated standard errors. In other words, it is hard to assess the accuracy of an estimate other than the mean.

Suppose for example, we want to compare the two groups in Table 2.1 by their medians rather than their means. The two medians are 94 for treatment and 46 for control, giving an estimated difference of 48, considerably more than the difference of the means. But how accurate are these medians? Answering such questions is where the bootstrap, and other computer-based techniques, come in. The remainder of this chapter gives a brief preview of the bootstrap estimate of standard error, a method which will be fully discussed in succeeding chapters.

Suppose we observe independent data points $x_1, x_2, \cdots, x_n$, for convenience denoted by the vector $\mathbf{x} = (x_1, x_2, \cdots, x_n)$, from which we compute a statistic of interest $s(\mathbf{x})$. For example the data might be the $n = 9$ control group observations in Table 2.1, and $s(\mathbf{x})$ might be the sample mean.

The bootstrap estimate of standard error, invented by Efron in 1979, looks completely different than (2.2), but in fact it is closely related, as we shall see. A *bootstrap sample* $\mathbf{x}^* = (x_1^*, x_2^*, \cdots, x_n^*)$ is obtained by randomly sampling $n$ times, with replacement, from the original data points $x_1, x_2, \cdots, x_n$. For instance, with $n = 7$ we might obtain $\mathbf{x}^* = (x_5, x_7, x_5, x_4, x_7, x_3, x_1)$.

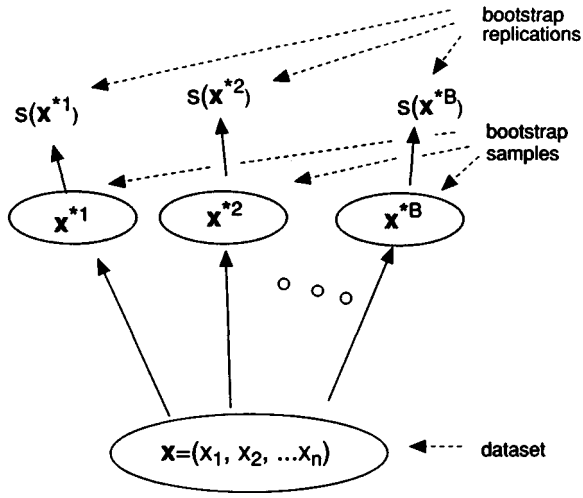



Figure 2.1. *Schematic of the bootstrap process for estimating the standard error of a statistic* $s(\mathbf{x})$. *B bootstrap samples are generated from the original data set. Each bootstrap sample has n elements, generated by sampling with replacement n times from the original data set. Bootstrap replicates* $s(\mathbf{x}^{*1}), s(\mathbf{x}^{*2}), \ldots s(\mathbf{x}^{*B})$ *are obtained by calculating the value of the statistic* $s(\mathbf{x})$ *on each bootstrap sample. Finally, the standard deviation of the values* $s(\mathbf{x}^{*1}), s(\mathbf{x}^{*2}), \ldots s(\mathbf{x}^{*B})$ *is our estimate of the standard error of* $s(\mathbf{x})$.

Figure 2.1 is a schematic of the bootstrap process. The bootstrap algorithm begins by generating a large number of independent bootstrap samples $\mathbf{x}^{*1}, \mathbf{x}^{*2}, \cdots, \mathbf{x}^{*B}$, each of size $n$. Typical values for $B$, the number of bootstrap samples, range from 50 to 200 for standard error estimation. Corresponding to each bootstrap sample is a *bootstrap replication* of $s$, namely $s(\mathbf{x}^{*b})$, the value of the statistic $s$ evaluated for $\mathbf{x}^{*b}$. If $s(\mathbf{x})$ is the sample median, for instance, then $s(\mathbf{x}^*)$ is the median of the bootstrap sample. The bootstrap estimate of standard error is the standard deviation of the bootstrap replications,

$$\widehat{\mathrm{se}}_{\mathrm{boot}} = \left\{ \sum_{b=1}^{B} [s(\mathbf{x}^{*b}) - s(\cdot)]^2/(B-1) \right\}^{\frac{1}{2}}, \tag{2.3}$$

where $s(\cdot) = \sum_{b=1}^{B} s(\mathbf{x}^{*b})/B$. Suppose $s(\mathbf{x})$ is the mean $\bar{x}$. In this

Table 2.2. *Bootstrap estimates of standard error for the mean and median; treatment group, mouse data, Table 2.1. The median is less accurate (has larger standard error) than the mean for this data set.*

| B: | 50 | 100 | 250 | 500 | 1000 | $\infty$ |
|---|---|---|---|---|---|---|
| mean: | 19.72 | 23.63 | 22.32 | 23.79 | 23.02 | 23.36 |
| median: | 32.21 | 36.35 | 34.46 | 36.72 | 36.48 | 37.83 |

case, standard probability theory tells us (Problem 2.5) that as $B$ gets very large, formula (2.3) approaches

$$\{\sum_{i=1}^{n}(x_i - \bar{x})^2/n^2\}^{\frac{1}{2}}. \tag{2.4}$$

This is almost the same as formula (2.2). We could make it exactly the same by multiplying definition (2.3) by the factor $[n/(n-1)]^{\frac{1}{2}}$, but there is no real advantage in doing so.

Table 2.2 shows bootstrap estimated standard errors for the mean and the median, for the treatment group mouse data of Table 2.1. The estimated standard errors settle down to limiting values as the number of bootstrap samples $B$ increases. The limiting value 23.36 for the mean is obtained from (2.4). The formula for the limiting value 37.83 for the standard error of the median is quite complicated: see Problem 2.4 for a derivation.

We are now in a position to assess the precision of the difference in medians between the two groups. The bootstrap procedure described above was applied to the control group, producing a standard error estimate of 11.54 based on $B = 100$ replications ($B = \infty$ gave 9.73). Therefore, using $B = 100$, the observed difference of 48 has an estimated standard error of $\sqrt{36.35^2 + 11.54^2} = 38.14$, and hence is $48/38.14 = 1.26$ standard errors greater than zero. This is larger than the observed difference in means, but is still insignificant.

For most statistics we don't have a formula for the limiting value of the standard error, but in fact no formula is needed. Instead we use the numerical output of the bootstrap program, for some convenient value of $B$. We will see in Chapters 6 and 19, that $B$ in the range 50 to 200 usually makes $\widehat{\text{se}}_{\text{boot}}$ a good standard error

estimator, even for estimators like the median. It is easy to write a bootstrap program that works for any computable statistic $s(\mathbf{x})$, as shown in Chapters 6 and the Appendix. With these programs in place, the data analyst is free to use any estimator, no matter how complicated, with the assurance that he or she will also have a reasonable idea of the estimator's accuracy. The price, a factor of perhaps 100 in increased computation, has become affordable as computers have grown faster and cheaper.

Standard errors are the simplest measures of statistical accuracy. Later chapters show how bootstrap methods can assess more complicated accuracy measures, like biases, prediction errors, and confidence intervals. Bootstrap confidence intervals add another factor of 10 to the computational burden. The payoff for all this computation is an increase in the statistical problems that can be analyzed, a reduction in the assumptions of the analysis, and the elimination of the routine but tedious theoretical calculations usually associated with accuracy assessment.

## 2.1 Problems

2.1† Suppose that the mouse survival times were expressed in weeks instead of days, so that the entries in Table 2.1 were all divided by 7.

(a) What effect would this have on $\bar{x}$ and on its estimated standard error (2.2)? Why does this make sense?

(b) What effect would this have on the ratio of the difference $\bar{x} - \bar{y}$ to its estimated standard error?

2.2 Imagine the treatment group in Table 2.1 consisted of $R$ repetitions of the data actually shown, where $R$ is a positive integer. That is, the treatment data consisted of $R$ 94's, $R$ 197's, etc. What effect would this have on the estimated standard error (2.2)?

2.3 It is usually true that the error of a statistical estimator decreases at a rate of about 1 over the square root of the sample size. Does this agree with the result of Problem 2.2?

2.4 Let $x_{(1)} < x_{(2)} < x_{(3)} < x_{(4)} < x_{(5)} < x_{(6)} < x_{(7)}$ be an ordered sample of size $n = 7$. Let $\mathbf{x}^*$ be a bootstrap sample, and $s(\mathbf{x}^*)$ be the corresponding bootstrap replication of the median. Show that

(a) $s(\mathbf{x}^*)$ equals one of the original data values $x_{(i)}$, $i = 1, 2, \cdots, 7$.

(b) † $s(\mathbf{x}^*)$ equals $x_{(i)}$ with probability

$$p(i) = \sum_{j=0}^{3} \{\text{Bi}(j; n, \frac{i-1}{n}) - \text{Bi}(j; n, \frac{i}{n})\}, \tag{2.5}$$

where $\text{Bi}(j; n, p)$ is the binomial probability $\binom{n}{j} p^j (1-p)^{n-j}$. [The numerical values of $p(i)$ are .0102, .0981, .2386, .3062, .2386, .0981, .0102. These values were used to compute $\widehat{\text{se}}_{\text{boot}}\{\text{median}\} = 37.83$, for $B = \infty$, Table 2.2.]

2.5 Apply the weak law of large numbers to show that expression (2.3) approaches expression (2.4) as $n$ goes to infinity.

† Indicates a difficult or more advanced problem.

CHAPTER 3

# Random samples and probabilities

## 3.1 Introduction

Statistics is the theory of accumulating information, especially information that arrives a little bit at a time. A typical statistical situation was illustrated by the mouse data of Table 2.1. No one mouse provides much information, since the individual results are so variable, but seven, or nine mice considered together begin to be quite informative. Statistical theory concerns the best ways of extracting this information. Probability theory provides the mathematical framework for statistical inference. This chapter reviews the simplest probabilistic model used to model random data: the case where the observations are a random sample from a single unknown population, whose properties we are trying to learn from the observed data.

## 3.2 Random samples

It is easiest to visualize random samples in terms of a finite population or "universe" $\mathcal{U}$ of individual units $U_1, U_2, \cdots, U_N$, any one of which is equally likely to be selected in a single random draw. The population of units might be all the registered voters in an area undergoing a political survey, all the men that might conceivably be selected for a medical experiment, all the high schools in the United States, etc. The individual units have properties we would like to learn, like a political opinion, a medical survival time, or a graduation rate. It is too difficult and expensive to examine every unit in $\mathcal{U}$, so we select for observation a random sample of manageable size.

A *random sample of size* $n$ is defined to be a collection of $n$

units $u_1, u_2, \cdots, u_n$ selected at random from $\mathcal{U}$. In principle the sampling process goes as follows: a random number device independently selects integers $j_1, j_2, \cdots, j_n$, each of which equals any value between 1 and $N$ with probability $1/N$. These integers determine which members of $\mathcal{U}$ are selected to be in the random sample, $u_1 = U_{j_1}, u_2 = U_{j_2}, \cdots, u_n = U_{j_n}$. In practice the selection process is seldom this neat, and the population $\mathcal{U}$ may be poorly defined, but the conceptual framework of random sampling is still useful for understanding statistical inference. (The methodology of good experimental design, for example the random assignment of selected units to Treatment or Control groups as was done in the mouse experiment, helps make random sampling theory more applicable to real situations like that of Table 2.1.)

Our definition of random sampling allows a single unit $U_i$ to appear more than once in the sample. We could avoid this by insisting that the integers $j_1, j_2, \cdots, j_n$ be distinct, called "sampling without replacement." It is a little simpler to allow repetitions, that is to "sample with replacement", as in the previous paragraph. If the size $n$ of the random sample is much smaller than the population size $N$, as is usually the case, the probability of sample repetitions will be small anyway. See Problem 3.1. Random sampling always means sampling *with* replacement in what follows, unless otherwise stated.

Having selected a random sample $u_1, u_2, \cdots, u_n$, we obtain one or more measurements of interest for each unit. Let $x_i$ indicate the measurements for unit $u_i$. The *observed data* are the collection of measurements $x_1, x_2, \cdots, x_n$. Sometimes we will denote the observed data $(x_1, x_2, \cdots, x_n)$ by the single symbol $\mathbf{x}$.

We can imagine making the measurements of interest on every member $U_1, U_2, \cdots, U_N$ of $\mathcal{U}$, obtaining values $X_1, X_2, \cdots, X_N$. This would be called a census of $U$.

The symbol $\mathcal{X}$ will denote the census of measurements $(X_1, X_2, \cdots, X_N)$. We will also refer to $\mathcal{X}$ as the population of measurements, or simply the population, and call $\mathbf{x}$ a random sample of size $n$ from $\mathcal{X}$. In fact, we usually can't afford to conduct a census, which is why we have taken a random sample. The goal of statistical inference is to say what we have learned about the population $\mathcal{X}$ from the observed data $\mathbf{x}$. In particular, we will use the bootstrap to say how accurately a statistic calculated from $x_1, x_2, \cdots, x_n$ (for instance the sample median) estimates the corresponding quantity for the whole population.

Table 3.1. *The law school data. A random sample of size $n = 15$ was taken from the collection of $N = 82$ American law schools participating in a large study of admission practices. Two measurements were made on the entering classes of each school in 1973: LSAT, the average score for the class on a national law test, and GPA, the average undergraduate grade-point average for the class.*

| School | LSAT | GPA | School | LSAT | GPA |
|---|---|---|---|---|---|
| 1 | 576 | 3.39 | 9 | 651 | 3.36 |
| 2 | 635 | 3.30 | 10 | 605 | 3.13 |
| 3 | 558 | 2.81 | 11 | 653 | 3.12 |
| 4 | 578 | 3.03 | 12 | 575 | 2.74 |
| 5 | 666 | 3.44 | 13 | 545 | 2.76 |
| 6 | 580 | 3.07 | 14 | 572 | 2.88 |
| 7 | 555 | 3.00 | 15 | 594 | 2.96 |
| 8 | 661 | 3.43 | | | |

Table 3.1 shows a random sample of size $n = 15$ drawn from a population of $N = 82$ American law schools. What is actually shown are two measurements made on the entering classes of 1973 for each school in the sample: LSAT, the average score of the class on a national law test, and GPA, the average undergraduate grade point average achieved by the members of the class. In this case the measurement $x_i$ on $u_i$, the $i$th member of the sample, is the pair

$$x_i = (\text{LSAT}_i, \text{GPA}_i) \qquad i = 1, 2, \cdots, 15.$$

The observed data $x_1, x_2, \cdots, x_n$ is the collection of 15 pairs of numbers shown in Table 3.1.

This example is an artificial one because the census of data $X_1, X_2, \cdots, X_{82}$ was actually made. In other words, LSAT and GPA are available for the entire population of $N = 82$ schools. Figure 3.1 shows the census data and the sample data. Table 3.2 gives the entire population of $N$ measurements.

In a real statistical problem, like that of Table 3.1, we would see only the sample data, from which we would be trying to infer the properties of the population. For example, consider the 15 LSAT scores in the observed sample. These have mean 600.27 with estimated standard error 10.79, based on the data in Table 3.1 and formula (2.2). There is about a 68% chance that the true LSAT

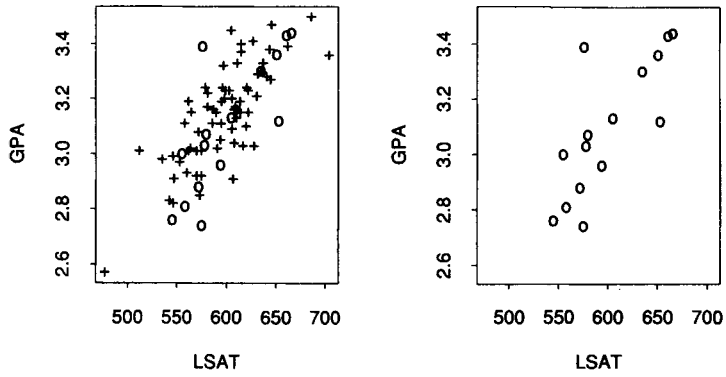



Figure 3.1. *The left panel is a scatterplot of the (LSAT, GPA) data for all $N = 82$ law schools; circles indicate the $n = 15$ data points comprising the "observed sample" of Table 3.1. The right panel shows only the observed sample. In problems of statistical inference, we are trying to infer the situation on the left from the picture on the right.*

mean, the mean for the entire population from which the observed data was sampled, lies in the interval $600.27 \pm 10.79$.

We can check this result, since we are dealing with an artificial example for which the complete population data are known. The mean of all 82 LSAT values is 597.55, lying nicely within the predicted interval $600.27 \pm 10.79$.

## 3.3 Probability theory

Statistical inference concerns learning from experience: we observe a random sample $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ and wish to infer properties of the complete population $\mathcal{X} = (X_1, X_2, \cdots, X_N)$ that yielded the sample. Probability theory goes in the opposite direction: from the composition of a population $\mathcal{X}$ we deduce the properties of a random sample $\mathbf{x}$, and of statistics calculated from $\mathbf{x}$. Statistical inference as a mathematical science has been developed almost exclusively in terms of probability theory. Here we will review briefly

Table 3.2. *The population of measurements (LSAT,GPA), for the universe of 82 law schools. The data in Table 3.1 was sampled from this population. The +'s indicate the sampled schools.*

| school | LSAT | GPA | school | LSAT | GPA | school | LSAT | GPA |
|---|---|---|---|---|---|---|---|---|
| 1 | 622 | 3.23 | 28 | 632 | 3.29 | 56 | 641 | 3.28 |
| 2 | 542 | 2.83 | 29 | 587 | 3.16 | 57 | 512 | 3.01 |
| 3 | 579 | 3.24 | 30 | 581 | 3.17 | 58 | 631 | 3.21 |
| 4+ | 653 | 3.12 | 31+ | 605 | 3.13 | 59 | 597 | 3.32 |
| 5 | 606 | 3.09 | 32 | 704 | 3.36 | 60 | 621 | 3.24 |
| 6+ | 576 | 3.39 | 33 | 477 | 2.57 | 61 | 617 | 3.03 |
| 7 | 620 | 3.10 | 34 | 591 | 3.02 | 62 | 637 | 3.33 |
| 8 | 615 | 3.40 | 35+ | 578 | 3.03 | 62 | 572 | 3.08 |
| 9 | 553 | 2.97 | 36+ | 572 | 2.88 | 64 | 610 | 3.13 |
| 10 | 607 | 2.91 | 37 | 615 | 3.37 | 65 | 562 | 3.01 |
| 11 | 558 | 3.11 | 38 | 606 | 3.20 | 66 | 635 | 3.30 |
| 12 | 596 | 3.24 | 39 | 603 | 3.23 | 67 | 614 | 3.15 |
| 13+ | 635 | 3.30 | 40 | 535 | 2.98 | 68 | 546 | 2.82 |
| 14 | 581 | 3.22 | 41 | 595 | 3.11 | 69 | 598 | 3.20 |
| 15+ | 661 | 3.43 | 42 | 575 | 2.92 | 70+ | 666 | 3.44 |
| 16 | 547 | 2.91 | 43 | 573 | 2.85 | 71 | 570 | 3.01 |
| 17 | 599 | 3.23 | 44 | 644 | 3.38 | 72 | 570 | 2.92 |
| 18 | 646 | 3.47 | 45+ | 545 | 2.76 | 73 | 605 | 3.45 |
| 19 | 622 | 3.15 | 46 | 645 | 3.27 | 74 | 565 | 3.15 |
| 20 | 611 | 3.33 | 47+ | 651 | 3.36 | 75 | 686 | 3.50 |
| 21 | 546 | 2.99 | 48 | 562 | 3.19 | 76 | 608 | 3.16 |
| 22 | 614 | 3.19 | 49 | 609 | 3.17 | 77 | 595 | 3.19 |
| 23 | 628 | 3.03 | 50+ | 555 | 3.00 | 78 | 590 | 3.15 |
| 24 | 575 | 3.01 | 51 | 586 | 3.11 | 79+ | 558 | 2.81 |
| 25 | 662 | 3.39 | 52+ | 580 | 3.07 | 80 | 611 | 3.16 |
| 26 | 627 | 3.41 | 53+ | 594 | 2.96 | 81 | 564 | 3.02 |
| 27 | 608 | 3.04 | 54 | 594 | 3.05 | 82+ | 575 | 2.74 |
| | | | 55 | 560 | 2.93 | | | |

some fundamental concepts of probability, including probability distributions, expectations, and independence.

As a first example, let $x$ represent the outcome of rolling a fair die so $x$ is equally likely to be $1, 2, 3, 4, 5$, or 6. We write this in probability notation as

$$\text{Prob}\{x = k\} = 1/6 \qquad \text{for} \quad k = 1, 2, 3, 4, 5, 6. \tag{3.1}$$

A random quantity like $x$ is often called a *random variable.*

Probabilities are idealized or theoretical proportions. We can imagine a universe $\mathcal{U} = \{U_1, U_2, \cdots, U_N\}$ of possible rolls of the

die, where $U_j$ completely describes the physical act of the $j$th roll, with corresponding results $\mathcal{X} = (X_1, X_2, \cdots, X_N)$. Here $N$ might be very large, or even infinite. The statement $\text{Prob}\{x = 5\} = 1/6$ means that a randomly selected member of $\mathcal{X}$ has a $1/6$ chance of equaling 5, or more simply that $1/6$ of the members of $\mathcal{X}$ equal 5. Notice that probabilities, like proportions, can never be less than 0 or greater than 1.

For convenient notation define the *frequencies* $f_k$,

$$f_k = \text{Prob}\{x = k\}, \tag{3.2}$$

so the fair die has $f_k = 1/6$ for $k = 1, 2, \cdots, 6$. The *probability distribution of a random variable* $x$, which we will denote by $F$, is any complete description of the probabilistic behavior of $x$. $F$ is also called the probability distribution of the population $\mathcal{X}$. Here we can take $F$ to be the vector of frequencies

$$F = (f_1, f_2, \cdots, f_6) = (1/6, 1/6, \cdots, 1/6). \tag{3.3}$$

An *unfair* die would be one for which $F$ did not equal $(1/6, 1/6, \cdots, 1/6)$.

Note: In many books, the symbol $F$ is used for the cumulative probability distribution function $F(x_0) = \text{Prob}\{x \leq x_0\}$ for $-\infty < x_0 < \infty$. This is an equally valid description of the probabilistic behavior of $x$, but it is only convenient for the case where $x$ is a real number. We will also be interested in cases where $x$ is a vector, as in Table 3.1, or an even more general object. This is the reason for defining $F$ as *any* description of $x$'s probabilities, rather than the specific description in terms of the cumulative probabilities. When no confusion can arise, in later chapters we use symbols like $F$ and $G$ to represent cumulative distribution functions.

Some probability distributions arise so frequently that they have received special names. A random variable $x$ is said to have the *binomial distribution* with size $n$ and probability of success $p$, denoted

$$x \sim \text{Bi}(n, p), \tag{3.4}$$

if its frequencies are

$$f_k = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for} \quad k = 0, 1, 2, \cdots, n. \tag{3.5}$$

Here $n$ is a positive integer, $p$ is a number between 0 and 1, and $\binom{n}{k}$ is the binomial coefficient $n!/[k!(n-k)!]$. Figure 3.2 shows the

distribution $F = (f_0, f_1, \cdots, f_n)$ for $x \sim \text{Bi}(n,p)$, with $n = 25$ and $p = .25, .50$, and $.90$. We also write $F = \text{Bi}(n,p)$ to indicate situation (3.4).

Let $A$ be a set of integers. Then the probability that $x$ takes a value in $A$, or more simply the *probability of* $A$, is

$$\text{Prob}\{x \in A\} = \text{Prob}\{A\} = \sum_{k \in A} f_k. \tag{3.6}$$

For example if $A = \{1, 3, 5, \cdots, 25\}$ and $x \sim \text{Bi}(25,p)$, then $\text{Prob}\{A\}$ is the probability that a binomial random variable of size 25 and probability of success $p$ equals an odd integer. Notice that since $f_k$ is the theoretical proportion of times $x$ equals $k$, the sum $\sum_{k \in A} f_k = \text{Prob}\{A\}$ is the theoretical proportion of times $x$ takes its value in $A$.

The *sample space* of $x$, denoted $\mathcal{S}_x$, is the collection of possible values $x$ can have. For a fair die, $\mathcal{S}_x = \{1, 2, \cdots, 6\}$, while $\mathcal{S}_x = \{0, 1, 2, \cdots, n\}$ for a $\text{Bi}(n,p)$ distribution. By definition, $x$ occurs in $\mathcal{S}_x$ every time, that is, with theoretical proportion 1, so

$$\text{Prob}\{\mathcal{S}_x\} = \sum_{k \in \mathcal{S}_x} f_k = 1. \tag{3.7}$$

For any probability distribution on the integers the frequencies $f_j$ are nonnegative numbers summing to 1.

In our examples so far, the sample space $\mathcal{S}_x$ has been a subset of the integers. One of the convenient things about probability distributions is that they can be defined on quite general spaces. Consider the law school data of Figure 3.1. We might take $\mathcal{S}_x$ to be the positive quadrant of the plane,

$$\mathcal{S}_x = \mathcal{R}^{2+} = \{(y,z), y > 0, z > 0\}. \tag{3.8}$$

(This includes values like $x = (10^6, 10^9)$, but it doesn't hurt to let $\mathcal{S}_x$ be too big.) For a subset $A$ of $\mathcal{S}_x$, we would still write $\text{Prob}\{A\}$ to indicate the probability that $x$ occurs in $A$.

For example, we could take

$$A = \{(y,z) : 0 < y < 600, 0 < z < 3.0\}. \tag{3.9}$$

A law school $x \in A$ if its 1973 entering class had LSAT less than 600 and GPA less than 3.0. In this case we happen to know the complete population $\mathcal{X}$; it is the 82 points indicated on the left panel of Figure 3.1 and in Table 3.2. Of these, 16 are in $A$, so

$$\text{Prob}\{A\} = 16/82 = .195. \tag{3.10}$$

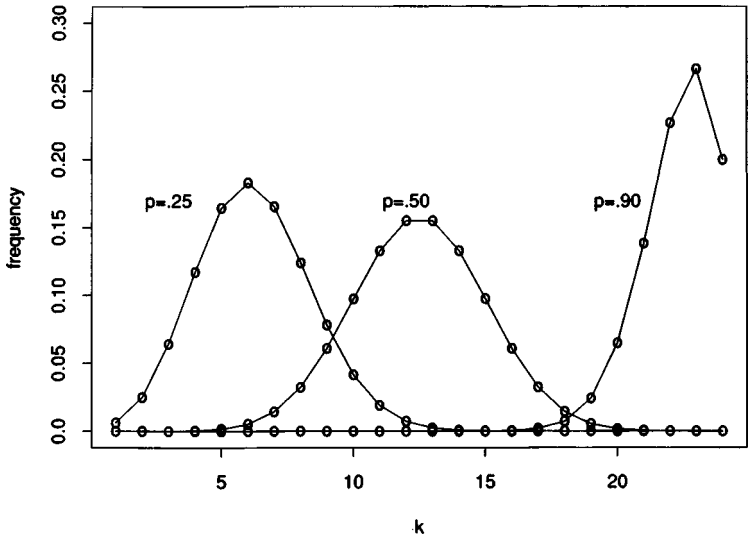



Figure 3.2. *The frequencies $f_0, f_1, \cdots, f_n$ for the binomial distributions $Bi(n,p)$, $n = 25$ and $p = .25, .50$, and .90. The points have been connected by lines to enhance visibility.*

Here the idealized proportion Prob$\{A\}$ is an actual proportion. Only in cases where we have a complete census of the population is it possible to directly evaluate probabilities as proportions.

The probability distribution $F$ of $x$ is still defined to be any complete description of $x$'s probabilities. In the law school example, $F$ can be described as follows: for any subset $A$ of $\mathcal{S}_x = \mathcal{R}^{2+}$,

$$\text{Prob}\{x \in A\} = \#\{X_j \in A\}/82, \tag{3.11}$$

where $\#\{X_j \in A\}$ is the number of the 82 points in the left panel of Figure 3.1 that lie in $A$. Another way to say the same thing is that $F$ is a *discrete distribution* putting probability (or frequency) 1/82 on each of the indicated 82 points.

Probabilities can be defined continuously, rather than discretely as in (3.6) or (3.11). The most famous example is the *normal* (or *Gaussian*, or *bell-shaped*) distribution. A real-valued random variable $x$ is defined to have the normal distribution with mean $\mu$ and

variance $\sigma^2$, written

$$x \sim N(\mu, \sigma^2) \quad \text{or} \quad F = N(\mu, \sigma^2), \tag{3.12}$$

if

$$\text{Prob}\{x \in A\} = \int_A \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \tag{3.13}$$

for any subset $A$ of the real line $\mathcal{R}^1$. The integral in (3.13) is over the values of $x \in A$.

There are higher dimensional versions of the normal distribution, which involve taking integrals similar to (3.13) over multidimensional sets $A$. We won't need continuous distributions for development of the bootstrap (though they will appear later in some of the applications) and will avoid mathematical derivations based on calculus. As we shall see, one of the main incentives for the development of the bootstrap is the desire to substitute computer power for theoretical calculations involving special distributions.

The *expectation* of a real-valued random variable $x$, written $\text{E}(x)$, is its average value, where the average is taken over the possible outcomes of $x$ weighted according to its probability distribution $F$. Thus

$$\text{E}(x) = \sum_{x=0}^{n} x \binom{n}{x} p^x (1-p)^x \quad \text{for} \quad x \sim \text{Bi}(n, p), \tag{3.14}$$

and

$$\text{E}(x) = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \quad \text{for} \quad x \sim N(\mu, \sigma^2). \tag{3.15}$$

It is not difficult to show that $\text{E}(x) = np$ for $x \sim \text{Bi}(n, p)$, and $\text{E}(x) = \mu$ for $x \sim N(\mu, \sigma^2)$. (See Problems 3.6 and 3.7.)

We sometimes write the expectation as $\text{E}_F(x)$, to indicate that the average is taken with respect to the distribution $F$.

Suppose $r = g(x)$ is some function of the random variable $x$. Then $\text{E}(r)$, the expectation of $r$, is the theoretical average of $g(x)$ weighted according to the probability distribution of $x$. For example if $x \sim N(\mu, \sigma^2)$ and $r = x^3$, then

$$\text{E}(r) = \int_{-\infty}^{\infty} x^3 \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx. \tag{3.16}$$

Probabilities are a special case of expectations. Let $A$ be a subset

of $\mathcal{S}_x$, and take $r = I_{\{x \in A\}}$ where $I_{\{x \in A\}}$ is the *indicator function*

$$I_{\{x \in A\}} = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}. \tag{3.17}$$

Then $\mathrm{E}(r)$ equals $\mathrm{Prob}\{x \in A\}$, or equivalently

$$\mathrm{E}(I_{\{x \in A\}}) = \mathrm{Prob}\{x \in A\}. \tag{3.18}$$

For example if $x \sim N(\mu, \sigma^2)$, then

$$\begin{aligned} \mathrm{E}(r) &= \int_{-\infty}^{\infty} I_{\{x \in A\}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx \\ &= \int_A \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2} dx, \end{aligned} \tag{3.19}$$

which is $\mathrm{Prob}\{x \in A\}$ according to (3.13).

The notion of an expectation as a theoretical average is very general, and includes cases where the random variable $x$ is not real-valued. In the law school situation, for instance, we might be interested in the expectation of the ratio of LSAT and GPA. Writing $x = (y, z)$ as in (3.8), then $r = y/z$, and the expectation of $r$ is

$$\mathrm{E}(\mathrm{LSAT/GPA}) = \frac{1}{82} \sum_{j=1}^{82} (y_j / z_j) \tag{3.20}$$

where $x_j = (y_j, z_j)$ is the $j$th point in Table 3.2. Numerical evaluation of (3.20) gives $\mathrm{E}(\mathrm{LSAT/GPA}) = 190.8$.

Let $\mu_x = \mathrm{E}_F(x)$, for $x$ a real-valued random variable with distribution $F$. The *variance* of $x$, indicated by $\sigma_x^2$ or just $\sigma^2$, is defined to be the expected value of $y = (x - \mu)^2$. In other words, $\sigma^2$ is the theoretical average squared distance of a random variable $x$ from its expectation $\mu_x$,

$$\sigma_x^2 = \mathrm{E}_F(x - \mu_x)^2. \tag{3.21}$$

The variance of $x \sim N(\mu, \sigma^2)$ equals $\sigma^2$; the variance of $x \sim \mathrm{Bi}(n, p)$ equals $np(1-p)$, see Problem 3.9. The *standard deviation* of a random variable is defined to be the square root of its variance.

Two random variables $y$ and $z$ are said to be *independent* if

$$\mathrm{E}[g(y)h(z)] = \mathrm{E}[g(y)]\mathrm{E}[h(z)] \tag{3.22}$$

for all functions $g(y)$ and $h(z)$. Independence is well named: (3.22) implies that the random outcome of $y$ doesn't affect the random outcome of $z$, and vice-versa.

To see this, let $B$ and $C$ be subsets of $\mathcal{S}_y$ and $\mathcal{S}_z$ respectively, the sample spaces of $y$ and $z$, and take $g$ and $h$ to be the indicator functions $g(y) = I_{\{y \in B\}}$ and $h(z) = I_{\{z \in C\}}$. Notice that

$$I_{\{y \in B\}} I_{\{z \in C\}} = \begin{cases} 1 & \text{if } y \in B \text{ and } z \in C \\ 0 & \text{otherwise.} \end{cases} \tag{3.23}$$

So $I_{\{y \in B\}} I_{\{z \in C\}}$ is the indicator function of the intersection $\{y \in B\} \cap \{z \in C\}$. Then by (3.18) and the independence definition (3.22),

$$\begin{aligned} \text{Prob}\{(y,z) \in B \cap C\} &= \text{E}(I_{\{y \in B\}} I_{\{z \in C\}}) = \text{E}(I_{\{y \in B\}})\text{E}(I_{\{z \in C\}}) \\ &= \text{Prob}\{y \in B\}\text{Prob}\{z \in C\}. \end{aligned} \tag{3.24}$$

Looking at Figure 3.1, we can see that (3.24) does *not* hold for the law school example, see Problem 3.10, so LSAT and GPA are not independent.

Whether or not $y$ and $z$ are independent, expectations follow the simple addition rule

$$\text{E}[g(y) + h(z)] = \text{E}[g(y)] + \text{E}[h(z)]. \tag{3.25}$$

In general,

$$\text{E}[\sum_{i=1}^{n} g_i(x_i)] = \sum_{i=1}^{n} \text{E}[g_i(x_i)] \tag{3.26}$$

for any functions $g_i$ of any $n$ random variables $x_1, x_2, \cdots, x_n$.

Random sampling with replacement guarantees independence: if $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ is a random sample of size $n$ from a population $\mathcal{X}$, then all $n$ observations $x_i$ are identically distributed and mutually independent of each other. In other words, all of the $x_i$ have the same probability distribution $F$, and

$$\begin{aligned} &\text{E}_F[g_1(x_1)g_2(x_2), \cdots, g_n(x_n)] = \\ &\quad \text{E}_F[g_1(x_1)]\text{E}_F[g_2(x_2)] \cdots \text{E}_F[g_n(x_n)] \end{aligned} \tag{3.27}$$

for any functions $g_1, g_2, \cdots, g_n$. (This is almost a definition of what random sampling means.) We will write

$$F \to (x_1, x_2, \cdots, x_n) \tag{3.28}$$

to indicate that $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ is a random sample of size $n$ from a population with probability distribution $F$. This is sometimes written as

$$x_i \overset{\text{i.i.d.}}{\sim} F \qquad i = 1, 2, \cdots, n, \tag{3.29}$$

where i.i.d. stands for independent and identically distributed.

## 3.4 Problems

3.1 A random sample of size $n$ is taken *with* replacement from a population of size $N$. Show that the probability of having no repetitions in the sample is given by the product

$$\prod_{j=0}^{n-1}(1 - \frac{j}{N}).$$

3.2 Why might you suspect that the sample of 15 law schools in Table (3.1) was obtained by sampling without replacement, rather than with replacement?

3.3 The mean GPA for all 82 law schools is 3.13. How does this compare with the mean GPA for the observed sample of 15 law schools in Table 3.1? Is this difference compatible with the estimated standard error (2.2)?

3.4 Denote the mean and standard deviation of a set of numbers $X_1, X_2, \cdots, X_N$ by $\overline{X}$ and $S$ respectively, where

$$\overline{X} = \sum_{j=1}^{N} X_j/N \qquad S = \{\sum_{j=1}^{N}(X_j - \overline{X})^2/N\}^{1/2}.$$

(a) A sample $x_1, x_2, \cdots, x_n$ is selected from $X_1, X_2, \cdots, X_N$ by random sampling with replacement. Denote the standard deviation of the sample average $\bar{x} = \sum_{i=1}^{n} x_i/n$, usually called *the standard error* of $\bar{x}$, by $\text{se}(\bar{x})$. Use a basic result of probability theory to show that

$$\text{se}(\bar{x}) = \frac{S}{\sqrt{n}}.$$

(b) † Suppose instead that $x_1, x_2, \cdots, x_n$ is selected by random sampling *without* replacement (so we must have

$n \le N$), show that

$$\text{se}(\bar{x}) = \frac{S}{\sqrt{n}}\left[\frac{N-n}{N-1}\right]^{\frac{1}{2}}$$

(c) We see that sampling without replacement gives a smaller standard error for $\bar{x}$. Proportionally how much smaller will it be in the case of the law school data?

3.5 Given a random sample $x_1, x_2, \cdots, x_n$, the *empirical probability* of a set $A$ is defined to be the proportion of the sample in $A$, written

$$\widehat{\text{Prob}}\{A\} = \#\{x_i \in A\}/n. \tag{3.30}$$

(a) Find $\widehat{\text{Prob}}\{A\}$ for the data in Table 3.1, with $A$ as given in (3.9).

(b) The standard error of an empirical probability is $[\text{Prob}\{A\} \cdot (1 - \text{Prob}\{A\})/n]^{1/2}$. How many standard errors is $\widehat{\text{Prob}}\{A\}$ from $\text{Prob}\{A\}$, given in (3.10)?

3.6 A very simple probability distribution $F$ puts probability on only two outcomes, 0 or 1, with frequencies

$$f_0 = 1 - p, \quad f_1 = p. \tag{3.31}$$

This is called the *Bernoulli* distribution. Here $p$ is a number between 0 and 1. If $x_1, \cdots, x_n$ is a random sample from $F$, then elementary probability theory tells us that the sum

$$s = x_1 + x_2 + \cdots + x_n \tag{3.32}$$

has the binomial distribution (3.5),

$$s \sim \text{Bi}(n, p). \tag{3.33}$$

(a) Show that the empirical probability (3.30) satisfies

$$n \cdot \widehat{\text{Prob}}\{A\} \sim \text{Bi}(n, \text{Prob}\{A\}). \tag{3.34}$$

Expression (3.34) can also be written as $\widehat{\text{Prob}}\{A\} \sim \text{Bi}(n, \text{Prob}\{A\})/n$.)

(b) Prove that if $x \sim \text{Bi}(n, p)$, then $\text{E}(x) = np$.

3.7 Without using calculus, give a symmetry argument to show that $\text{E}(x) = \mu$ for $x \sim N(\mu, \sigma^2)$.

3.8 Suppose that $y$ and $z$ are independent random variables, with variances $\sigma_y^2$ and $\sigma_z^2$.

(a) Show that the variance of $y + z$ is the sum of the variances

$$\sigma_{y+z}^2 = \sigma_y^2 + \sigma_z^2 \,. \tag{3.35}$$

(In general, the variance of the sum is the sum of the variances for independent random variables $x_1, x_2, \cdots, x_n$.)

(b) Suppose $F \to (x_1, x_2, \cdots, x_n)$ where the probability distribution $F$ has expectation $\mu$ and variance $\sigma^2$. Show that $\bar{x}$ has expectation $\mu$ and variance $\sigma^2/n$.

3.9 Use the results in Problems (3.6) and (3.8) to show that $\sigma_x^2 = np(1-p)$ for $x \sim \text{Bi}(n, p)$.

3.10 Forty-three of the 82 points in Table 3.1 have LSAT $< 600$; 17 of the 82 points have GPA $< 3.0$. Why do we know that LSAT and GPA are not independent?

3.11 In the discussion of random sampling, $j_1, j_2, \cdots, j_n$ were taken to be independent integers having a uniform distribution on the numbers $1, 2, \cdots, N$. That is, $j_1, j_2, \cdots, j_n$ is itself a random sample, say

$$F_{1:N} \to (j_1, j_2, \cdots, j_n), \tag{3.36}$$

where $F_{1:N}$ is the discrete distribution having frequencies $f_j = 1/N$, for $j = 1, 2, \cdots, N$. In practice, we depend on our computer's random number generator to give us (3.36). If (3.36) holds, then a random sample as defined in this chapter has the "i.i.d." property defined in (3.29). Give a brief argument why this is so.

† Indicates a difficult or more advanced problem.

CHAPTER 4

# The empirical distribution function and the plug-in principle

## 4.1 Introduction

Problems of statistical inference often involve estimating some aspect of a probability distribution $F$ on the basis of a random sample drawn from $F$. The empirical distribution function, which we will call $\hat{F}$, is a simple estimate of the entire distribution $F$. An obvious way to estimate some interesting aspect of $F$, like its mean or median or correlation, is to use the corresponding aspect of $\hat{F}$. This is the "plug-in principle." The bootstrap method is a direct application of the plug-in principle, as we shall see in Chapter 6.

## 4.2 The empirical distribution function

Having observed a random sample of size $n$ from a probability distribution $F$,

$$F \rightarrow (x_1, x_2, \cdots, x_n), \tag{4.1}$$

the *empirical distribution function* $\hat{F}$ is defined to be the discrete distribution that puts probability $1/n$ on each value $x_i$, $i = 1, 2, \cdots, n$. In other words, $\hat{F}$ assigns to a set $A$ in the sample space of $x$ its empirical probability

$$\widehat{\text{Prob}}\{A\} = \#\{x_i \in A\}/n, \tag{4.2}$$

the proportion of the observed sample $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ occurring in $A$. We will also write $\text{Prob}_{\hat{F}}\{A\}$ to indicate (4.2). The hat symbol "$\wedge$" always indicates quantities calculated from the observed data.

Table 4.1. *A random sample of 100 rolls of the die. The outcomes* $1, 2, 3, 4, 5, 6$ *occurred* $13, 19, 10, 17, 14, 27$ *times, respectively, so the empirical distribution is* $(.13, .19, .10, .17, .14, .27)$.

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 3 | 2 | 4 | 6 | 6 | 6 | 5 | 3 | 6 | 2 | 2 | 6 | 2 | 3 | 1 | 5 | 1 |
| 6 | 6 | 4 | 1 | 5 | 3 | 6 | 6 | 4 | 1 | 4 | 2 | 5 | 6 | 6 | 5 | 5 | 3 |
| 6 | 2 | 6 | 6 | 1 | 4 | 1 | 5 | 6 | 1 | 6 | 3 | 3 | 2 | 2 | 2 | 5 | 2 |
| 2 | 4 | 1 | 4 | 5 | 6 | 6 | 6 | 2 | 2 | 4 | 6 | 1 | 2 | 2 | 2 | 5 | 1 |
| 5 | 3 | 5 | 4 | 2 | 1 | 4 | 6 | 6 | 5 | 6 | 4 | 6 | 4 | 3 | 6 | 4 | 1 |
| 4 | 5 | 4 | 4 | 2 | 3 | 2 | 1 | 4 | 6 | | | | | | | | |

Consider the law school sample of size $n = 15$, shown in Table 3.1 and in the right panel of Figure 3.1. The empirical distribution $\hat{F}$ puts probability $1/15$ on each of the 15 data points. Five of the 15 points lie in the set $A = \{(y, z) : 0 < y < 600, 0 < z < 3.00\}$, so $\widehat{\text{Prob}}\{A\} = 5/15 \doteq .333$. Notice that we get a different empirical probability for the set $\{0 < y < 600, 0 < z \leq 3.00\}$, since one of the 15 data points has GPA $= 3.00$, LSAT $< 600$.

Table 4.1 shows a random sample of $n = 100$ rolls of a die: $x_1 = 6, x_2 = 3, x_3 = 2, \cdots, x_{100} = 6$. The empirical distribution $\hat{F}$ puts probability $1/100$ on each of the 100 outcomes. In cases like this, where there are repeated values, we can express $\hat{F}$ more economically as the vector of *observed frequencies* $\hat{f}_k$, $k = 1, 2, \cdots, 6$,

$$\hat{f}_k = \#\{x_i = k\}/n. \tag{4.3}$$

For the data in Table 4.1, $\hat{F} = (.13, .19, .10, .17, .14, .27)$.

An empirical distribution is a list of the values taken on by the sample $\mathbf{x} = (x_1, x_2, \cdots, x_n)$, along with the proportion of times each value occurs. Often each value occurring in the sample appears only once, as with the law data. Repetitions, as with the die of Table 4.1, allow the list to be shortened. In either case each of the $n$ data points $x_i$ is assigned probability $1/n$ by the empirical distribution.

Is it obvious that we have not lost information in going from the full data set $(x_1, x_2, \cdots, x_{100})$ in Table 4.1 to the reduced representation in terms of the frequencies? No, but it is true. It can be proved that the vector of observed frequencies $\hat{F} = (\hat{f}_1, \hat{f}_2, \cdots)$ is a *sufficient statistic* for the true distribution $F = (f_1, f_2, \cdots)$. This means that all of the information about $F$ contained in $\mathbf{x}$ is also contained in $\hat{F}$.

Table 4.2. *Rainfall data. The yearly rainfall, in inches, in Nevada City, California, 1873 through 1978. An example of time series data.*

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1870: | | | | 80 | 40 | 65 | 46 | 68 | 32 | 58 |
| 1880: | 60 | 61 | 60 | 45 | 48 | 63 | 44 | 66 | 39 | 35 |
| 1890: | 44 | 104 | 36 | 45 | 69 | 50 | 72 | 57 | 53 | 30 |
| 1900: | 40 | 56 | 55 | 46 | 46 | 72 | 50 | 68 | 71 | 37 |
| 1910: | 64 | 46 | 69 | 31 | 33 | 61 | 56 | 55 | 40 | 37 |
| 1920: | 40 | 34 | 60 | 54 | 52 | 20 | 49 | 43 | 62 | 44 |
| 1930: | 33 | 45 | 30 | 53 | 32 | 38 | 56 | 63 | 52 | 79 |
| 1940: | 30 | 62 | 75 | 70 | 60 | 34 | 54 | 51 | 35 | 53 |
| 1950: | 44 | 53 | 73 | 80 | 54 | 52 | 40 | 77 | 52 | 75 |
| 1960: | 42 | 43 | 39 | 54 | 70 | 40 | 73 | 41 | 75 | 43 |
| 1970: | 80 | 60 | 59 | 41 | 67 | 83 | 56 | 29 | 21 | |

The sufficiency theorem assumes that the data have been generated by random sampling from some distribution $F$. This is certainly *not* always true. For example the mouse data of Table 2.1 involve two probability distributions, one for Treatment and one for Control. Table 4.2 shows a *time-series* of 106 numbers: the annual rainfall in Nevada City, California from 1873 through 1978. We could calculate the empirical distribution $\hat{F}$ for this data set, but it would not include any of time series information, for example, if high numbers follow high numbers. Later, in Chapter 8, we will see how to apply bootstrap methods to situations like the rainfall data. For now we are restricting attention to data obtained by random sampling from a single distribution, the so-called *one-sample situation.* This is not as restrictive as it sounds. In the mouse data example, for instance, we can apply one-sample results separately to the Treatment and Control populations.

In applying statistical theory to real problems, the answers to questions of interest are usually phrased in terms of probability distributions. We might ask if the die giving the data in Table 4.1 is fair. This is equivalent to asking if the die's probability distribution $F$ equals $(1/6, 1/6, 1/6, 1/6, 1/6, 1/6)$. In the law school example, the question might be how correlated are LSAT and GPA. In terms of $F$, the distribution of $x = (y, z) = (\text{LSAT}, \text{GPA})$, this is

a question about the value of the *population correlation coefficient*

$$\text{corr}(y,z) = \frac{\sum_{j=1}^{82}(Y_j - \mu_y)(Z_j - \mu_z)}{[\sum_{j=1}^{82}(Y_j - \mu_y)^2 \sum_{j=1}^{82}(Z_j - \mu_z)^2]^{1/2}}, \tag{4.4}$$

where $(Y_j, Z_j)$ is the $j$th point in the law school population $\mathcal{X}$, and $\mu_y = \sum_{j=1}^{82} Y_j/82$, $\mu_z = \sum_{j=1}^{82} Z_j/82$.

When the probability distribution $F$ is known (i.e. when we have a complete census of the population $\mathcal{X}$), answering such questions involves no more than arithmetic. For the law school population, the census in Table 3.2 gives $\mu_y = 597.5$, $\mu_z = 3.13$, and

$$\text{corr}(y,z) = .761. \tag{4.5}$$

This is the original definition of "statistics." Usually we don't have a census. Then we need statistical inference, the more modern statistical theory for inferring properties of $F$ from a random sample $\mathbf{x}$.

If we had available only the law school sample of size 15, Table 3.1, we could estimate $\text{corr}(y,z)$ by the *sample correlation coefficient*

$$\widehat{\text{corr}}(y,z) = \frac{\sum_{i=1}^{15}(y_i - \hat{\mu}_y)(z_i - \hat{\mu}_z)}{[\sum_{i=1}^{15}(y_i - \hat{\mu}_y)^2 \sum_{i=1}^{15}(z_i - \hat{\mu}_z)^2]^{1/2}} \tag{4.6}$$

where $(y_i, z_i)$ is the $i$th point in Table 3.1, $i = 1, 2, \cdots, 15$, and $\hat{\mu}_y = \sum_{i=1}^{15} y_i/15$, $\hat{\mu}_z = \sum_{i=1}^{15} z_i/15$. Table 3.1 gives $\hat{\mu}_y = 600.3$, $\hat{\mu}_z = 3.09$, and

$$\widehat{\text{corr}}(y,z) = .776. \tag{4.7}$$

Here is another example of a plug-in estimate. Suppose we are interested in estimating the probability of a LSAT score greater than 600, that is

$$\theta = \frac{1}{82}\sum_{1}^{82} I_{\{Y_i > 600\}}. \tag{4.8}$$

Since 39 of the 82 LSAT scores exceed 600, $\theta = 39/82 \doteq 0.48$. The plug estimate of $\theta$ is

$$\hat{\theta} = \frac{1}{15}\sum_{1}^{15} I_{\{y_i > 600\}} \tag{4.9}$$

the sample proportion of LSAT scores above 600. Six of the 15 LSAT scores exceed 600, so $\hat{\theta} = 6/15 = 0.4$.

For the die of Table 4.1, we don't have census data but only the sample $\mathbf{x}$, so any questions about the fairness of the die must be answered by inference from the empirical frequencies

$$\hat{F} = (\hat{f}_1, \hat{f}_2, \cdots, \hat{f}_6) = (.13, .19, .10, .17, .14, .27). \tag{4.10}$$

Discussions of statistical inference are phrased in terms of *parameters* and *statistics*. A parameter is a function of the probability distribution $F$. A statistic is a function of the sample $\mathbf{x}$. Thus $\text{corr}(y, z)$, (4.4), is a parameter of $F$, while $\widehat{\text{corr}}(y, z)$, (4.6), is a statistic based on $\mathbf{x}$. Similarly $f_k$ is a parameter of $F$ in the die example, while $\hat{f}_k$ is a statistic, $k = 1, 2, 3, \cdots, 6$.

We will sometimes write parameters directly as functions of $F$, say

$$\theta = t(F). \tag{4.11}$$

This notation emphasizes that the value $\theta$ of the parameter is obtained by applying some numerical evaluation procedure $t(\cdot)$ to the distribution function $F$. For example if $F$ is a probability distribution in the real line, the expectation can be thought of as the parameter

$$\theta = t(F) = \text{E}_F(x). \tag{4.12}$$

Here $t(F)$ gives $\theta$ by the expectation process, that is, the average value of $x$ weighted according to $F$. For a given distribution $F$ such as $F = \text{Bi}(n, p)$ we can evaluate $t(F) = np$. Even if $F$ is unknown, the form of $t(F)$ tells us the functional mapping that inputs $F$ and outputs $\theta$.

## 4.3 The plug-in principle

The plug-in principle is a simple method of estimating parameters from samples. The *plug-in estimate* of a parameter $\theta = t(F)$ is defined to be

$$\hat{\theta} = t(\hat{F}). \tag{4.13}$$

In other words, we estimate the function $\theta = t(F)$ of the probability distribution $F$ by the same function of the empirical distribution $\hat{F}$, $\hat{\theta} = t(\hat{F})$. (Statistics like (4.13) that are used to estimate parameters are sometimes called *summary statistics*, as well as *estimates*

and *estimators.*)

We have already used the plug-in principle in estimating $f_k$ by $\hat{f}_k$, and in estimating $\text{corr}(y, z)$ by $\widehat{\text{corr}}(y, z)$. To see this, note that our law school population $F$ can be written as $F = (f_1, f_2, \ldots f_{82})$ where each $f_j$, the probability of the $j$th law school, has value $1/82$. This is the probability distribution on $\mathcal{X}$, the 82 law school pairs. The population correlation coefficient can be written as

$$\text{corr}(y, z) = \frac{\sum_{j=1}^{82} f_j(Y_j - \mu_y)(Z_j - \mu_z)}{[\sum_{j=1}^{82} f_j(Y_j - \mu_y)^2 \sum_{j=1}^{82} f_j(Z_j - \mu_z)^2]^{1/2}}, \quad (4.14)$$

where

$$\mu_y = \sum_{j=1}^{82} f_j Y_j, \quad \mu_z = \sum_{j=1}^{82} f_j Z_j. \quad (4.15)$$

Setting each $f_j = 1/82$ gives expression (4.4). Now for our sample $(x_1, x_2, \ldots x_{15})$, the sample frequency $\hat{f}_j$ is the proportion of sample points equal to $X_j$:

$$\hat{f}_j = \#\{x_i = X_j\}/15, \quad j = 1, 2, \ldots 82. \quad (4.16)$$

For the sample of Table 3.1, $\hat{f}_1 = 0, \hat{f}_2 = 0, \hat{f}_3 = 0, \hat{f}_4 = 1/15$ etc. Now plugging these values $\hat{f}_j$ into expressions (4.15) and (4.14) gives $\hat{\mu}_y, \hat{\mu}_z$ and $\widehat{\text{corr}}(y, z)$ respectively. That is, $\hat{\mu}_y, \hat{\mu}_z$ and $\widehat{\text{corr}}(y, z)$ are *plug-in* estimates of $\mu_y, \mu_z$ and $\text{corr}(y, z)$.

In general, the plug-in estimate of an expectation $\theta = \text{E}_F(x)$ is

$$\hat{\theta} = \text{E}_{\hat{F}}(x) = \frac{1}{n} \sum_{i=1}^{n} x_i = \bar{x}. \quad (4.17)$$

How good is the plug-in principle? It is usually quite good, if the only available information about $F$ comes from the sample $\mathbf{x}$. Under this circumstance $\hat{\theta} = t(\hat{F})$ cannot be improved upon as an estimator of $\theta = t(F)$, at least not in the usual asymptotic $(n \to \infty)$ sense of statistical theory. For example if $\hat{f}_k$ is the plug-in frequency estimate $\#\{x_i = k\}/n$, then

$$\hat{f}_k \sim \text{Bi}(n, f_k)/n \quad (4.18)$$

as in Problem 3.6. In this case the estimator $\hat{f}_k$ is *unbiased* for $f_k$, $\text{E}(\hat{f}_k) = f_k$, with variance $f_k(1 - f_k)/n$. This is the smallest possible variance for an unbiased estimator of $f_k$.

We will use the bootstrap to study the bias and standard error of the plug-in estimate $\hat{\theta} = t(\hat{F})$. The bootstrap's virtue is that it produces biases and standard errors in an automatic way, no matter how complicated the functional mapping $\theta = t(F)$ may be. We will see that the bootstrap itself is an application of the plug-in principle.

The plug-in principle is less good in situations where there is information about $F$ other than that provided by the sample $\mathbf{x}$. We might know, or assume, that $F$ is a member of a *parametric family*, like the family of multivariate normal distributions. Or we might be in a *regression* situation, where we have available a collection of random samples $\mathbf{x}(z)$ depending on a predictor variable $z$. Then even if we are only interested in $F_{z_0}$, the distribution function for some specific value $z_0$ of $z$, there may be information about $F_{z_0}$ in the other samples $\mathbf{x}(z)$, especially those for which $z$ is near $z_0$. Regression models are discussed in Chapters 7 and 9.

The plug-in principle and the bootstrap can be adopted to parametric families and to regression models. See Section 6.5 of Chapter 6 and Chapter 9. For the next few chapters we assume that we are in the situation where we have only the one random sample $\mathbf{x}$ from a completely unknown distribution $F$. This is called the one-sample *nonparametric* setup.

## 4.4 Problems

4.1 Say carefully why the plug-in estimate of the expectation of a real-valued random variable is $\bar{x}$, the sample average.

4.2 We would like to estimate the variance $\sigma_x^2$ of a real-valued random variable $x$, having observed a random sample $x_1, x_2, \cdots, x_n$. What is the plug-in estimate of $\sigma_x^2$?

4.3 (a) Show that the standard error of an empirical frequency $\hat{f}_k$ is $\sqrt{f_k(1-f_k)/n}$. (You can use the result in problem 3.5b.)

(b) Do you believe that the die used to generate Table 4.1 is fair?

4.4 Suppose a random variable $x$ has possible values $1, 2, 3, \cdots$. Let $A$ be a subset of the positive integers.

(a) Show that $\widehat{\text{Prob}}\{A\} = \sum_{k \in A} \hat{f}_k$.

(b) Compare problems 4.3a and 3.5b, and conclude that the observed frequencies $\hat{f}_k$ are not independent of each other.

(c) Say in words why the observed frequencies aren't independent.

CHAPTER 5

# Standard errors and estimated standard errors

## 5.1 Introduction

Summary statistics such as $\hat{\theta} = t(\hat{F})$ are often the first outputs of a data analysis. The next thing we want to know is the accuracy of $\hat{\theta}$. The bootstrap provides accuracy estimates by using the plug-in principle to estimate the standard error of a summary statistic. This is the subject of Chapter 6. First we will discuss estimation of the standard error of a mean, where the plug-in principle can be carried out explicitly.

## 5.2 The standard error of a mean

Suppose that $x$ is a real-valued random variable with probability distribution $F$. Let us denote the expectation and variance of $F$ by the symbols $\mu_F$ and $\sigma_F^2$ respectively,

$$\mu_F = \mathrm{E}_F(x), \qquad \sigma_F^2 = \mathrm{var}_F(x) = \mathrm{E}_F[(x - \mu_F)^2]. \tag{5.1}$$

These are the quantities called $\mu_x$ and $\sigma_x^2$ in Chapter 3. Here we are emphasizing the dependence on $F$. The alternative notation "$\mathrm{var}_F(x)$" for the variance, sometimes abbreviated to $\mathrm{var}(x)$, means the same thing as $\sigma_F^2$. In what follows we will sometimes write

$$x \sim (\mu_F, \sigma_F^2) \tag{5.2}$$

to indicate concisely the expectation and variance of $x$.

Now let $(x_1, \cdots, x_n)$ be a random sample of size $n$ from the distribution $F$. The mean of the sample $\bar{x} = \sum_{i=1}^n x_i/n$ has expectation $\mu_F$ and variance $\sigma_F^2/n$,

$$\bar{x} \sim (\mu_F, \sigma_F^2/n). \tag{5.3}$$

In other words, the expectation of $\bar{x}$ is the same as the expectation of a single $x$, but the variance of $\bar{x}$ is $1/n$ times the variance of $x$. See Problem 3.8b. This is the reason for taking averages: the larger $n$ is, the smaller $\text{var}(\bar{x})$ is, so bigger $n$ means a better estimate of $\mu_F$.

The *standard error* of the mean $\bar{x}$, written $\text{se}_F(\bar{x})$ or $\text{se}(\bar{x})$, is the square root of the variance of $\bar{x}$,

$$\text{se}_F(\bar{x}) = [\text{var}_F(\bar{x})]^{1/2} = \sigma_F/\sqrt{n}. \tag{5.4}$$

Standard error is a general term for the standard deviation of a summary statistic.[1] They are the most common way of indicating statistical accuracy. Roughly speaking, we expect $\bar{x}$ to be less than one standard error away from $\mu_F$ about 68% of the time, and less than two standard errors away from $\mu_F$ about 95% of the time.

These percentages are based on the *central limit theorem*. Under quite general conditions on $F$, the distribution of $\bar{x}$ will be approximately normal as $n$ gets large, which we can write as

$$\bar{x} \overset{\cdot}{\sim} N(\mu_F, \sigma_F^2/n). \tag{5.5}$$

The expectation $\mu_F$ and variance $\sigma_F^2/n$ in (5.5) are exact, only the normality being approximate. Using (5.5), a table of the normal distribution gives

$$\text{Prob}\{|\bar{x} - \mu_F| < \frac{\sigma_F}{\sqrt{n}}\} \doteq .683, \qquad \text{Prob}\{|\bar{x} - \mu_F| < \frac{2\sigma_F}{\sqrt{n}}\} \doteq .954, \tag{5.6}$$

as illustrated in Figure 5.1. One of the advantages of the bootstrap is that we do not have to rely entirely on the central limit theorem. Later we will see how to get accuracy statements like (5.6) directly from the data (see Chapters 12–14 on bootstrap confidence intervals). It will then be clear that (5.6), which is correct for large values of $n$, can sometimes be quite inaccurate for the sample size actually available. Keeping this in mind, it is still true that the standard error of an estimate usually gives a good idea of its accuracy.

A simple example shows the limitations of the central limit theorem approximation. Suppose that $F$ is a distribution that puts

[1] In some books, the term "standard error" is used to denote an estimated standard deviation, that is, an estimate of $\sigma_F$ based on the data. That differs from our usage of the term.

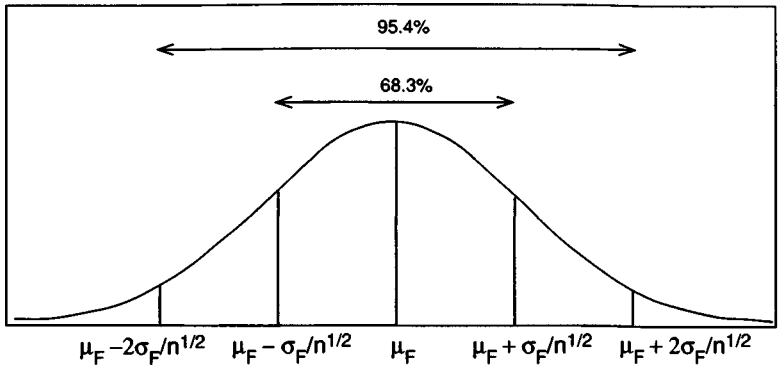



Figure 5.1. *For large values of $n$, the mean $\bar{x}$ of a random sample from $F$ will have an approximate normal distribution with mean $\mu_F$ and variance $\sigma_F^2/n$.*

probability on only two outcomes, 0 or 1, as in problem 3.6, say

$$\text{Prob}_F\{x = 1\} = p \qquad \text{and} \qquad \text{Prob}_F\{x = 0\} = 1 - p. \tag{5.7}$$

Here $p$ is a parameter of $F$, often called the probability of success, having a value between 0 and 1. A random sample $F \to (x_1, x_2, \cdots, x_n)$ can be thought of as $n$ independent flips of a coin having probability of success (or of "heads", or of $x = 1$) equaling $p$. Then the sum $s = \sum_{i=1}^{n} x_i$ is the number of successes in $n$ independent flips of the coin; $s$ has the binomial distribution (3.3),

$$s \sim \text{Bi}(n, p). \tag{5.8}$$

The average $\bar{x} = s/n$ equals $\hat{p}$, the plug-in estimate of $p$. Distribution (5.7) has $\mu_F = p$, $\sigma_F^2 = p(1-p)$, so (5.3) gives

$$\hat{p} \sim (p, p(1-p)/n) \tag{5.9}$$

for the mean and variance of $\hat{p}$. In other words, $\hat{p}$ is an unbiased estimate of $p$, $\text{E}(\hat{p}) = p$, with standard error

$$\text{se}(\hat{p}) = \left[\frac{p(1-p)}{n}\right]^{1/2}. \tag{5.10}$$

Figure 5.2 shows the central limit theorem working for the binomial distribution with $n = 25$, $p = .25$ and $p = .90$. (Problem 5.3 says what is actually plotted in Figure 5.2.) The central limit theorem gives a good approximation to the binomial distribution

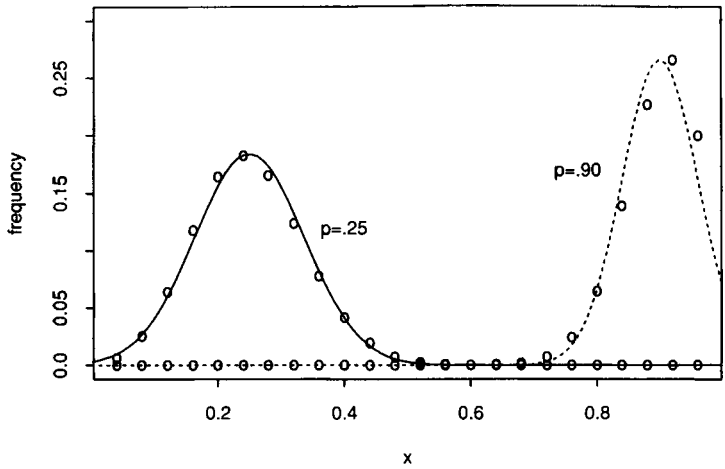



Figure 5.2. *Comparison of the binomial distribution with the normal distribution suggested by the central limit theorem; $n = 25$, $p = .25$ and $p = .90$. The smooth curves are the normal densities, see problem 5.3; circles indicate the binomial probabilities (3.5). The approximation is good for $p = .25$, but is somewhat off for $p = .90$.*

for $n = 25, p = .25$, but is somewhat less good for $n = 25, p = .9$.

## 5.3 Estimating the standard error of the mean

Suppose that we have in hand a random sample of numbers $F \to x_1, x_2, \cdots, x_n$, such as the $n = 9$ Control measurements for the mouse data of Table 2.1. We compute the estimate $\bar{x}$ for the expectation $\mu_F$, equaling 56.22 for the mouse data, and want to know the standard error of $\bar{x}$. Formula (5.4), $\text{se}_F(\bar{x}) = \sigma_F/\sqrt{n}$, involves the unknown distribution $F$ and so cannot be directly used.

At this point we can use the plug-in principle: we substitute $\hat{F}$ for $F$ in the formula $\text{se}_F(\bar{x}) = \sigma_F/\sqrt{n}$. The plug-in estimate of $\sigma_F = [\text{E}_F(x - \mu_F)^2]^{1/2}$ is

$$\hat{\sigma} = \sigma_{\hat{F}} = \{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\}^{1/2}, \tag{5.11}$$

since $\mu_{\hat{F}} = \bar{x}$ and $\mathrm{E}_{\hat{F}} g(x) = \frac{1}{n}\sum_{i=1}^{n} g(x_i)$ for any function $g$. This gives the *estimated standard error* $\widehat{\mathrm{se}}(\bar{x}) = \mathrm{se}_{\hat{F}}(\bar{x})$,

$$\widehat{\mathrm{se}}(\bar{x}) = \sigma_{\hat{F}}/\sqrt{n} = \{\sum_{i=1}^{n}(x_i - \bar{x})^2/n^2\}^{1/2}. \tag{5.12}$$

For the mouse Control group data, $\widehat{\mathrm{se}}(\bar{x}) = 13.33$.

Formula (5.12) is slightly different than the usual estimated standard error (2.2). That is because $\sigma_F$ is usually estimated by $\bar{\sigma} = \{\sum(x_i - \bar{x})^2/(n-1)\}^{1/2}$ rather than by $\hat{\sigma}$, (5.11). Dividing by $n-1$ rather than $n$ makes $\bar{\sigma}^2$ unbiased for $\sigma_F^2$. For most purposes $\hat{\sigma}$ is just as good as $\bar{\sigma}$ for estimating $\sigma_F$.

Notice that we have used the plug-in principle twice: first to estimate the expectation $\mu_F$ by $\mu_{\hat{F}} = \bar{x}$, and then to estimate the standard error $\mathrm{se}_F(\bar{x})$ by $\mathrm{se}_{\hat{F}}(\bar{x})$. The bootstrap estimate of standard error, which is the subject of Chapter 6, amounts to using the plug-in principle to estimate the standard error of an arbitrary statistic $\hat{\theta}$. Here we have seen that if $\hat{\theta} = \bar{x}$, then this approach leads to (almost) the usual estimate of standard error. As we will see, the advantage of the bootstrap is that it can be applied to virtually any statistic $\hat{\theta}$, not just the mean $\bar{x}$.

## 5.4 Problems

5.1 Formula (5.4) exemplifies a general statistical truth: most estimates of unknown quantities improve at a rate proportional to the square root of the sample size. Suppose that it were necessary to know $\mu_F$ for the mouse Control group with a standard error of no more than 3 days. How many more Control mice should be sampled?

5.2 State clearly why $\hat{p} = s/n$ is the plug-in estimate of $p$ for the binomial situation (5.8).

5.3 Figure 5.2 compares the function

$$\binom{n}{nx} p^{nx}(1-p)^{n(1-x)} \qquad \text{for} \qquad x = 0, 1/25, 2/25, \cdots, 1$$

with

$$\frac{1}{n}\frac{1}{\sqrt{2\pi p(1-p)/n}} \exp\{-\frac{1}{2}\Big[\frac{x-np}{\sqrt{np(1-p)}}\Big]^2\} \quad \text{for} \quad x \in [0,1].$$